

Pharmacology of Drugs Used in Migraine Headache



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Objectives

- ★ อธิบายกลไกการออกฤทธิ์และผลทางเภสัชวิทยาของยาที่เป็น abortive และ preventive treatment ใน migraine ได้
- ★ เปรียบเทียบความแตกต่างทางเภสัชวิทยาของ ergots, triptans และ gepants ได้
- ★ อธิบายหลักการทางเภสัชวิทยาของคำแนะนำที่สำคัญในการใช้ abortive และ preventive treatment ใน migraine ได้
- ★ อธิบายข้อควรระวัง ข้อห้ามใช้ ของยาที่ใช้ในการรักษา migraine ได้
- ★ ระบุข้อควรระวังการใช้ อาการไม่พึงประสงค์ และ drug interaction ที่สำคัญของ ergots และ triptans และ preventive treatment ได้

Features of migraine according to International Classification of Headache Disorder 3rd edition (ICHD-3)

Repeated (≥ 5) episodic headache (4-72 hrs)
with the following features:

★ At least two of :

- ◆ unilateral location
- ◆ pulsating quality (throbbing)
- ◆ moderate or severe pain intensity
- ◆ aggravation by or causing avoidance of routine physical activity (eg, walking or climbing stairs)

★ Any one of:

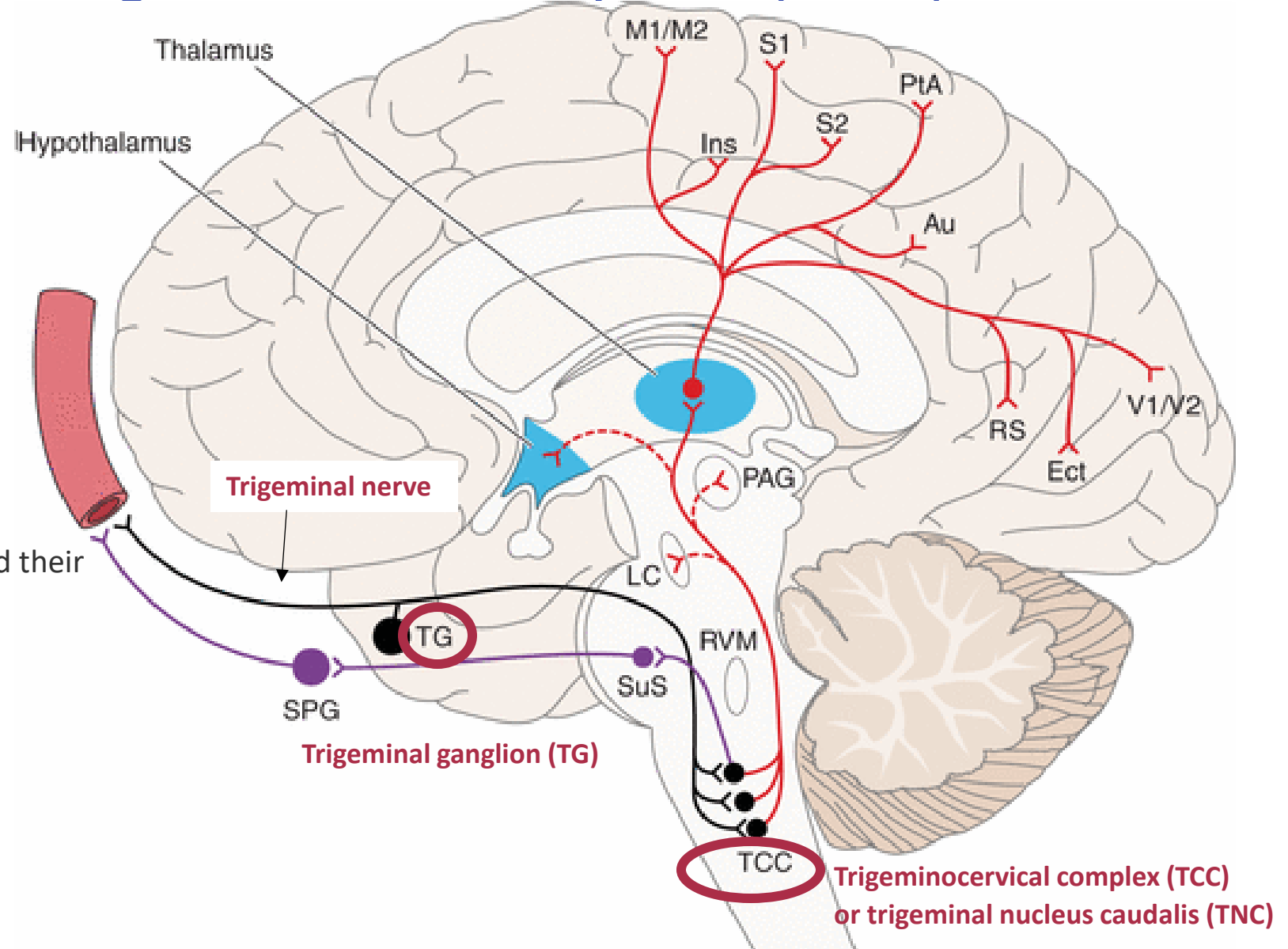
- ◆ Nausea and/or vomiting
- ◆ Photophobia and phonophobia

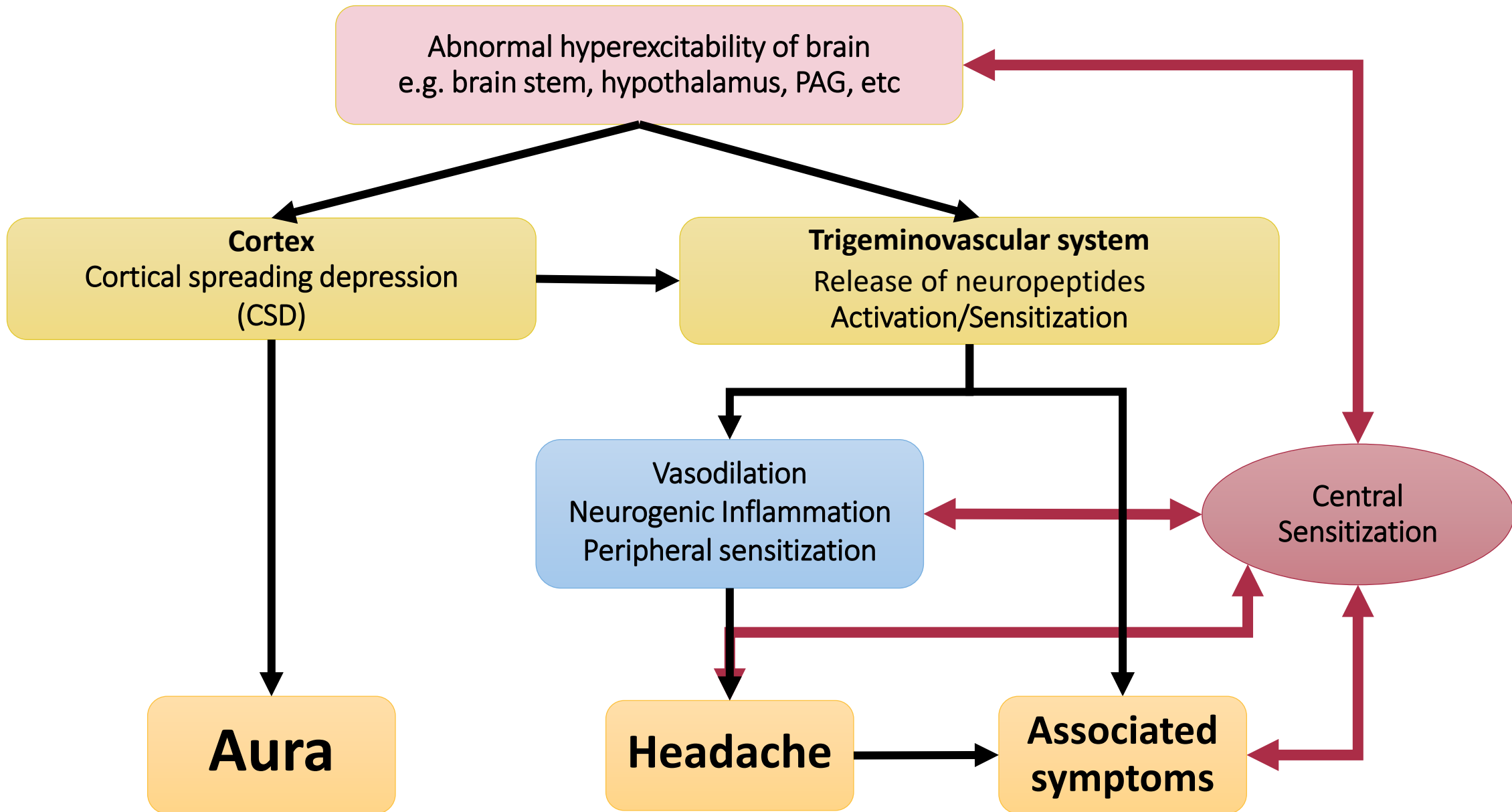
- Aura = Common migraine
+ Aura = Classic migraine

Anatomy of the trigeminovascular system (TGVS)

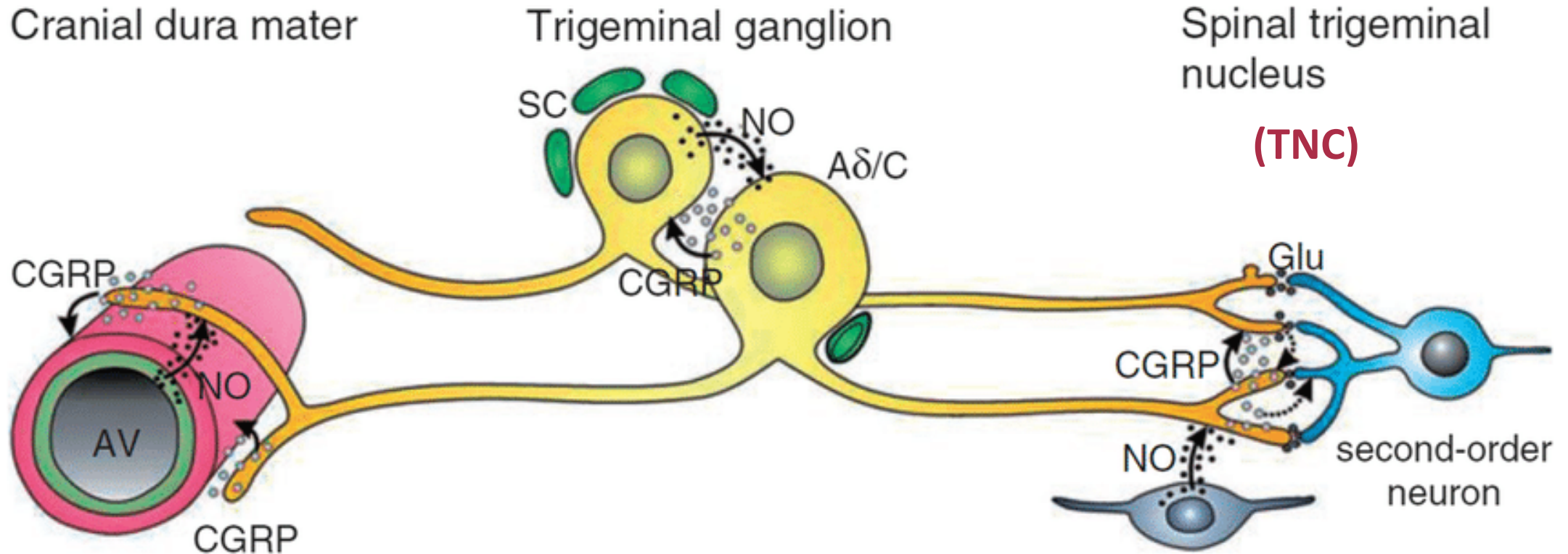
Cranial blood vessel: pain sensitive structure

- Arteries at base of the brain and their major branches
- Dural, meningeal arteries
- Venous sinuses





CGRP in the trigeminovascular system



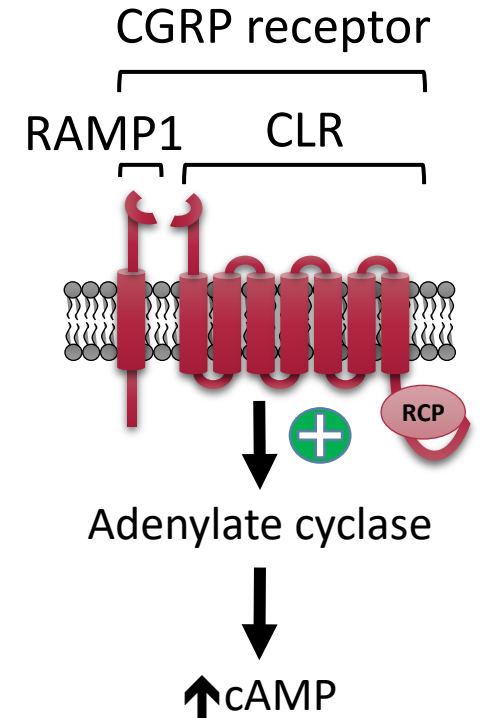
Calcitonin gene-related peptide (CGRP) and CGRP receptor

CGRP

- Multifunctional peptide (37 aa)
 - Autacoid
 - Neurotransmitter
- Potent vasodilator
 - 10x prostaglandins and 10-100x ACh, NO, substance P

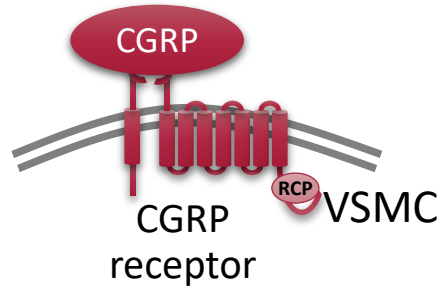
CGRP receptor

- GPCR coupled with Gs
- Complex structure: calcitonin receptor-like receptor (CLR) + receptor activity-modifying protein- 1 (RAMP-1)



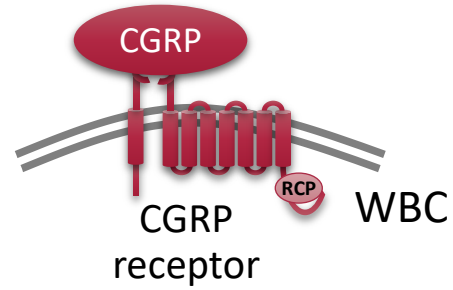
★ Both distributed widely in TGVS and other tissue e.g. CVS and GI tracts

Cranial vessel



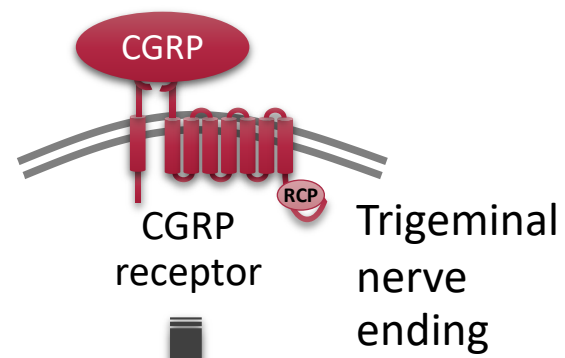
Vasodilation

Immune cell



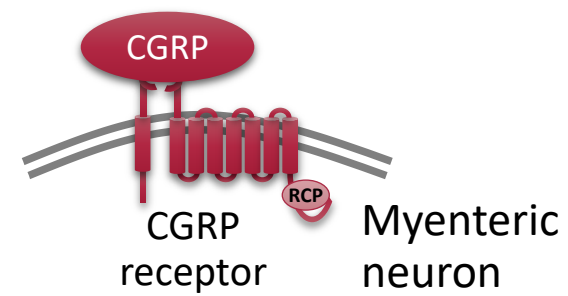
↑NO, PGE₂,
PGI₂ synthesis
Neurogenic inflammation

Neurons in pain pathway

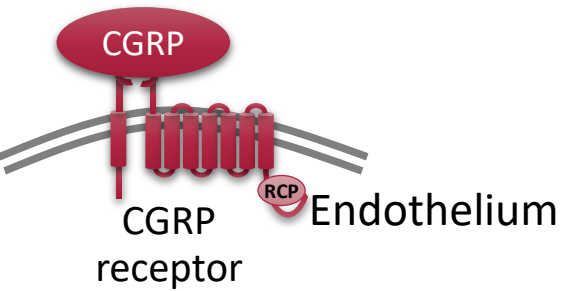


↑ excitability/
Peripheral sensitization

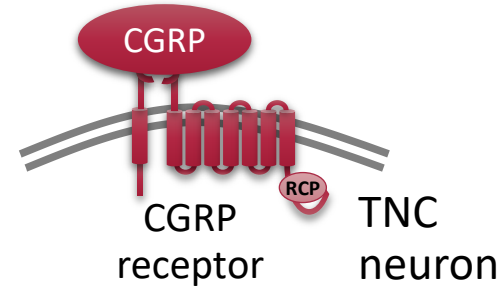
Neurons in GI tract



↑GI motility and
secretion



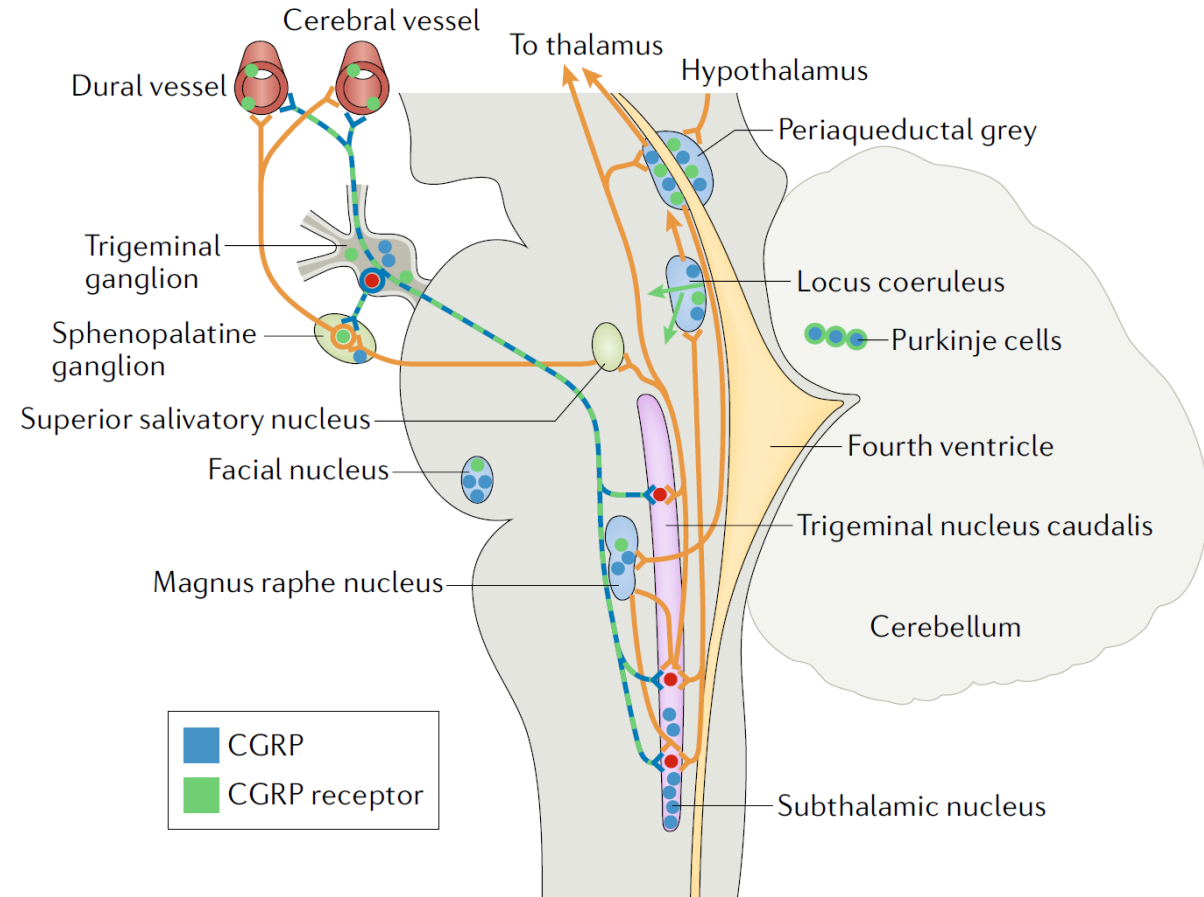
↑NO synthesis



↑ excitability/
Central sensitization

Roles of CGRP in headache pathophysiology

- ★ Increase neuronal excitability
 - ◈ Trigeminal nerve
 - ◈ TNC and associated areas
- ★ Vasodilation
 - ◈ Direct effect
 - ◈ Release of vasoactive substance esp. nitric oxide
- ★ Neurogenic inflammation
 - ◈ Recruitment and activation of immune cells to release proinflammatory mediators esp. PGE_2 , PGI_2



Drugs used in migraine headache

- ★ Abortive drugs
- ★ Preventive drugs

Drugs used in migraine headache: Abortive treatment

★ Specific

◈ 5-HT_{1B/1D/F} agonists

- ❖ Ergots: ergotamine, dihydroergotamine
- ❖ Triptans: sumatriptan, eletriptan, almotriptan, frovatriptan, naratriptan, rizatriptan, zolmitriptan
- ❖ Lasmiditan

◈ CGRP receptor antagonists (gepants)

◈ Combinations: Sumatriptan 50 mg and naproxen 250–500 mg

★ Non-specific

◈ Paracetamol, aspirin, ibuprofen, naproxen, diclofenac, etc.

Migraine-specific abortive therapy in Thailand

- ★ Ergotamine tartrate 1 mg + Caffeine 100 mg (Cafergot[®]) tablet



- ★ Sumatriptan (Imigran[®]) tablet 50 mg



- ★ Eletriptan (Relpax[®]) film-coated tablet 40 mg



- ★ Rimegepant (Nurtec[®]) orally disintegrating tablet 75 mg

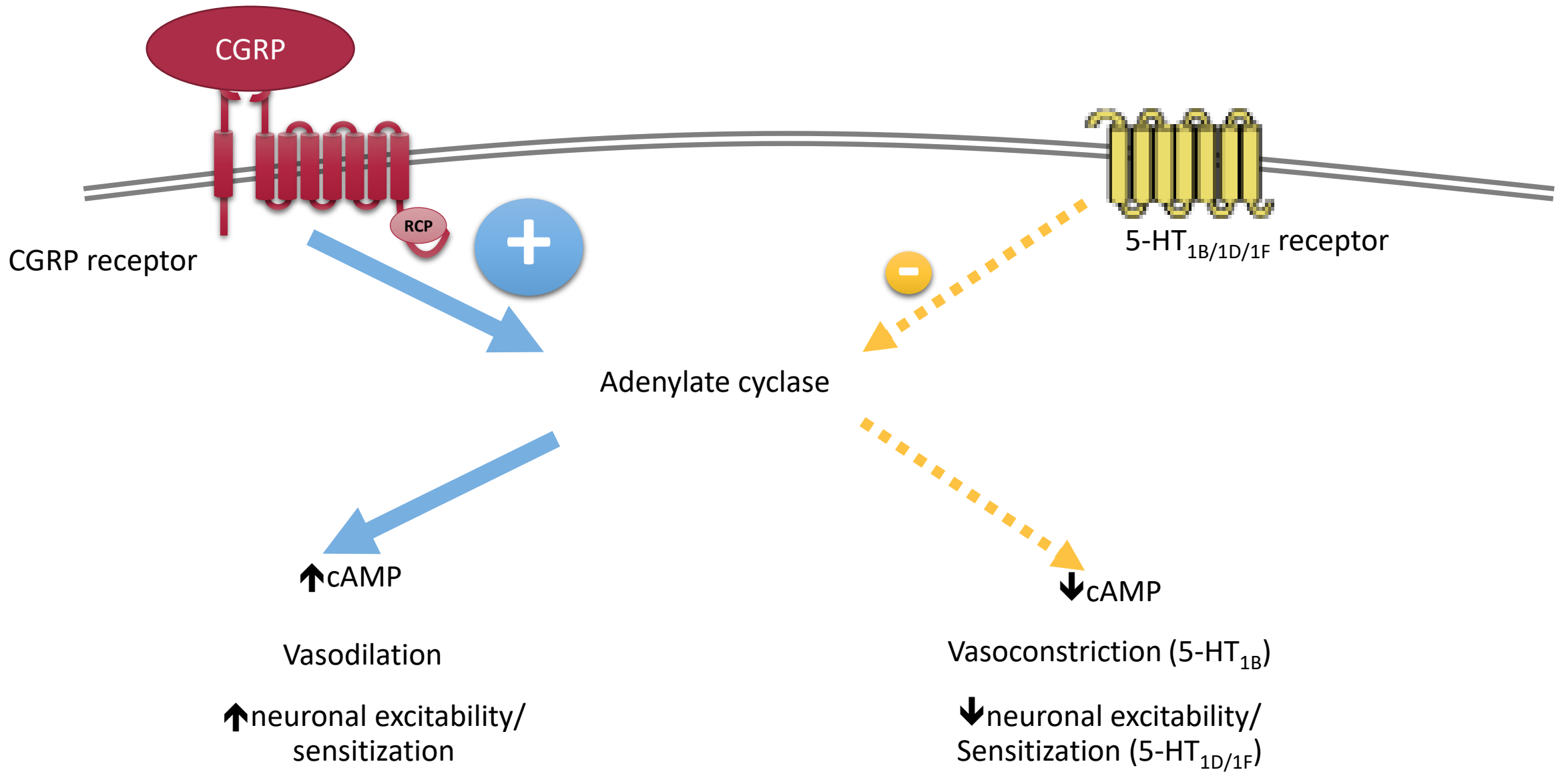


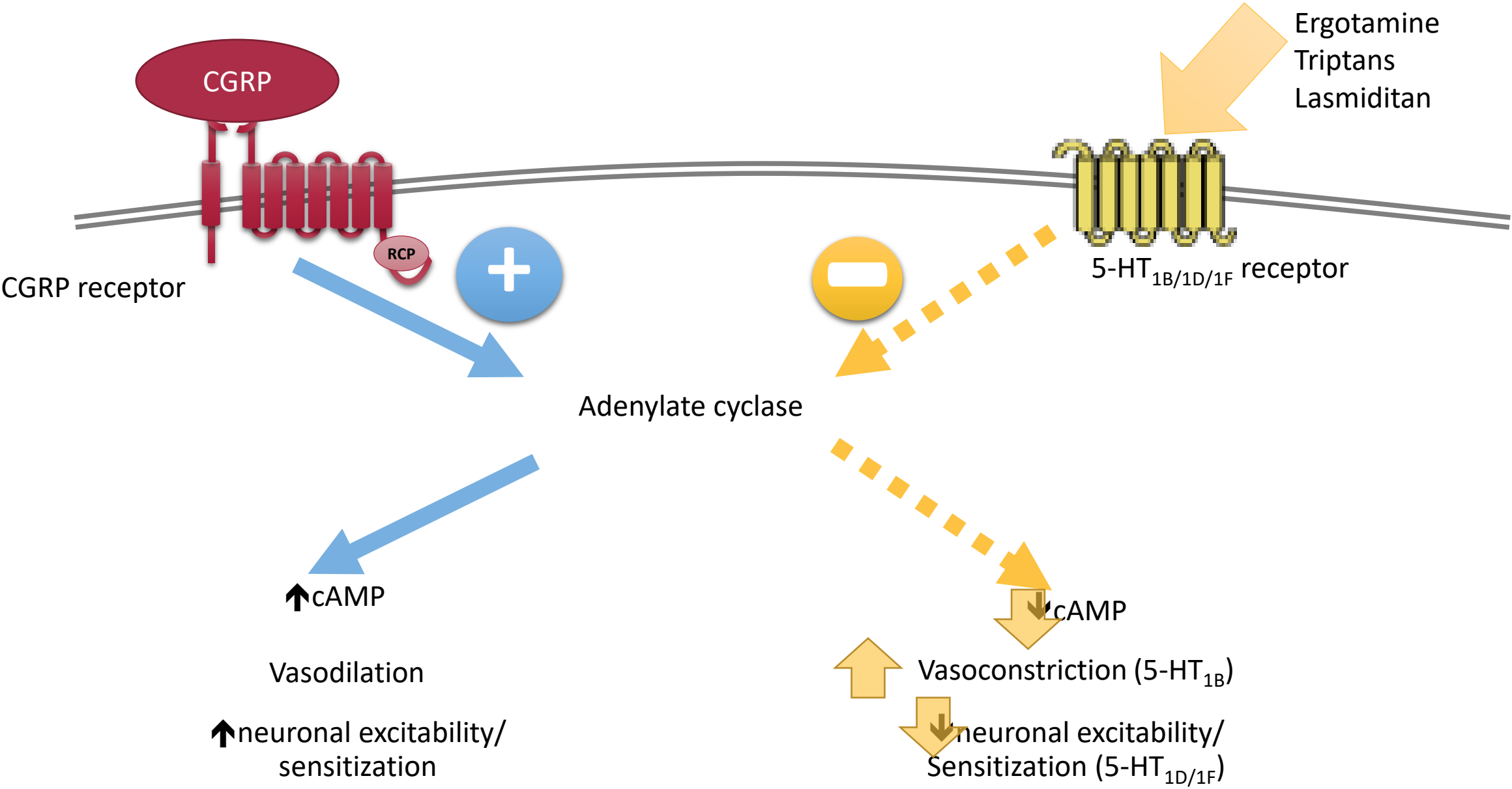
Mechanisms of action of 5-HT_{1B/1D/F} agonists

- ★ Ergotamine is nonselective to
 - ◈ Agonist of 5-HT_{1B/1D}
 - ◈ Agonist receptors α_1 , D₂, 5-HT_{2B}
- ★ Triptans are agonists of 5-HT_{1B/1D/1F}
 - ◈ More potent and selective than ergots
 - ◈ Less affinity to 5-HT_{1B}
 - ◈ More potent at 5-HT_{1F} than ergotamine
- ★ Lasmiditan is a potent selective of agonist 5-HT_{1F}

Distribution of 5-HT_{1B/1D/1F}

- ★ 5-HT_{1B} - trigeminal nerve, large cranial vessel and also other vascular beds e.g. coronary artery
- ★ 5-HT_{1D} – trigeminal nerve
- ★ 5-HT_{1F} – trigeminal nerve and TNC



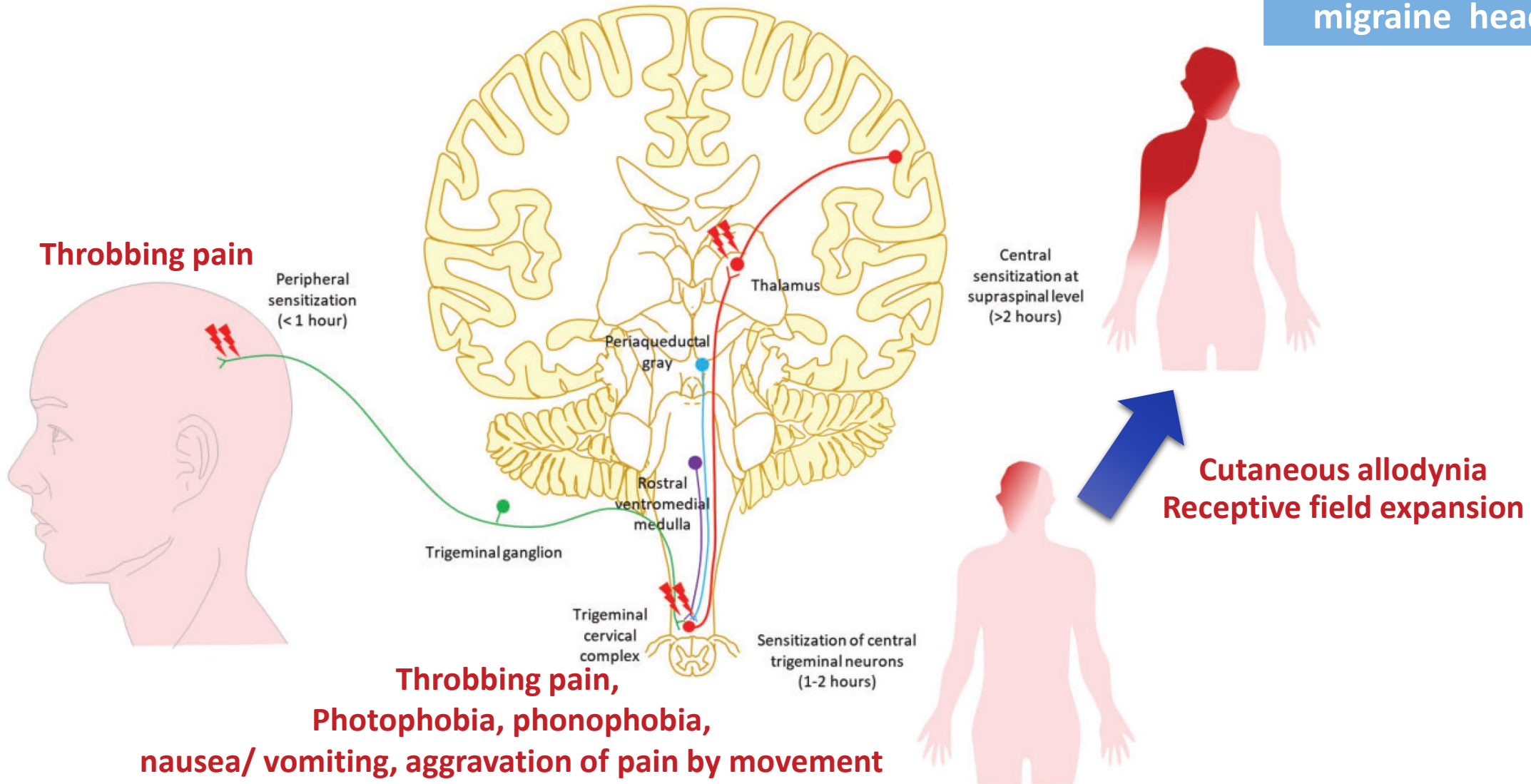


Pharmacokinetics and dosing of ergotamine and triptans

Parameter	Ergotamine	Sumatriptan	Eletriptan	Rimegepants
Bioavailability (oral)	1-5 %	15 %	50 %	64 %
Tmax (oral)	0.5-2 hr	2.5 hr	1-1.5 hr	1-1.5 hr
Half-life	3 hr	2.5 hr	3.6-6.9 hr	11 hr
Major metabolic pathway	CYP3A4	MAO-A	CYP3A4	CYP3A4
Usual initial dose	1-2 mg	50-100 mg	40 mg	75 mg
Interval to repeat dose	30 min	2 hr	2 hr	No
Maximum dose	6 mg/day 10 mg/week	200 mg /24hr	80 mg/24hr	75 mg/24 hr Or 75 mg every other day for prevention

General advice for all abortive drug

Migraine abortive therapy should be administered asap or within 1 hour after migraine headache onset



Contraindications/precautions of ergots

★ Contraindications

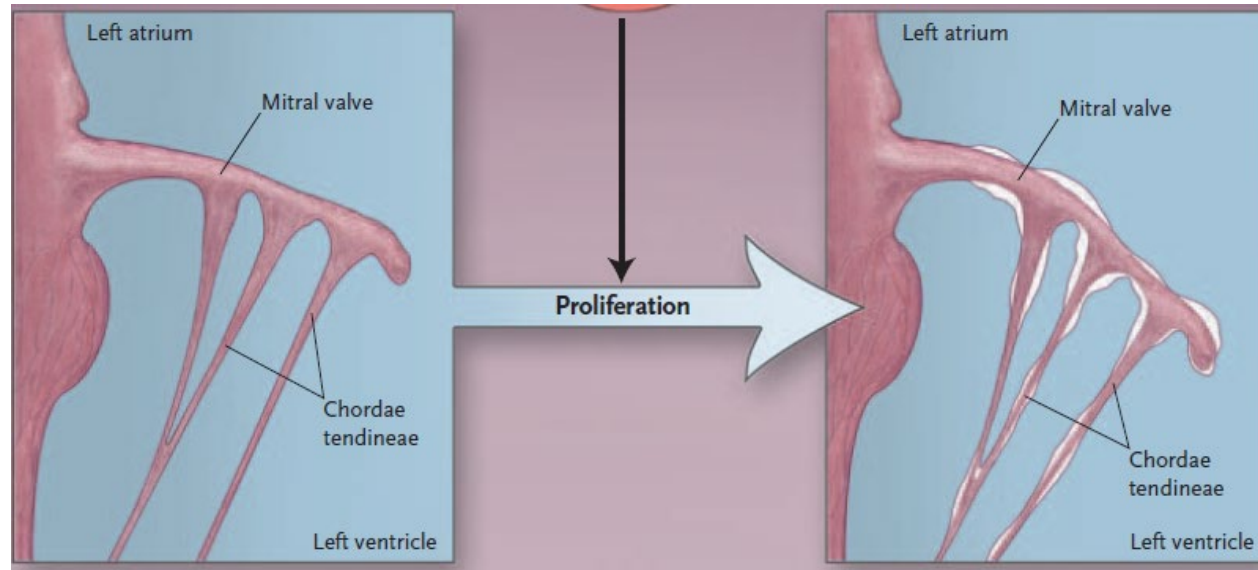
- ◈ Pregnancy
- ◈ Vascular diseases, uncontrollable hypertension
- ◈ Renal/liver dysfunction
- ◈ Familial hemiplegic migraine, basilar migraine

★ Precautions

- ◈ Concomitant use with CYP3A4 inhibitors or inducers
- ◈ DM, peripheral arterial disease

Inappropriate use of ergots

- ★ Overdose - > Ergotism
- ★ Long-term use
 - ◈ Tolerance leading to rebound headaches or medication-overused headache
 - ◈ Valvular/retroperitoneal/pleural fibrosis



Ergotism

★ Ergotism is a fatal condition that occurs from systemic vasoconstriction following ergot ingestion

- ◇ Abdominal and chest pain
- ◇ Cold, numbness, pain or tingling of extremities
- ◇ Cyanosis and possible gangrene
- ◇ Headache, psychosis
- ◇ Nausea, cramping, diarrhea

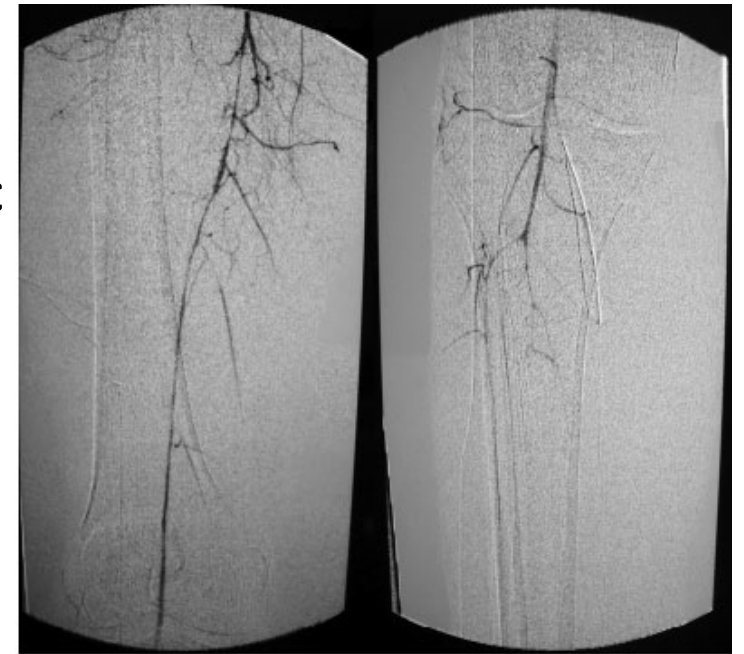


Figure 1 - Arteriography of right lower limb showing signs of spasm of the superficial femoral artery and fibular and posterior tibial arteries without opacification



Important drug-drug interactions of ergotamine

★ Ergotamine/eletriptan + CYP3A4 inhibitors

- ◆ Ergotamine/eletriptan should not be used within 3 days after a potent CYP3A4 inhibitor
- ◆ May lead to fatal ergotism

HIV antivirals: darunavir, indinavir, lopinavir, nelfinavir, ritonavir, saquinavir

Macrolide antibiotics: clarithromycin, erythromycin (except azitromycin)

Azole antifungals: itraconazole, ketoconazole, fluconazole

Other: Imatinib, fluvoxamine

★ Ergotamine/eletriptan/sumatriptan + drugs that increase serotonin level → serotonin syndrome

- ◆ Avoid coadministration of triptans or ergot-type medications within 24 hr

Triptans

- ★ The most common triptan side effects are known as 'triptan sensations' and they include
 - ◊ Paraesthesias
 - ◊ Flushing
 - ◊ Tingling
 - ◊ Neck pain
 - ◊ Mild transient chest pressure
- ★ Contraindicated in patients with vascular diseases, uncontrolled hypertension

Overuse of ergots, triptans and ditans

- ★ Tolerance
- ★ Withdrawal or rebound headache
- ★ Medication overuse headache

CGRP receptor antagonists

★ History

- ◆ Olcegepant and telcagepant had been developed until early 2000s
- ◆ Terminated due to hepatotoxicity

★ Abortive treatment

- ◆ Rimegepant orally disintegrating tablet (Nurtec[®] ODT)
- ◆ Ubrogepant tablet (Ubrelvy[®])
- ◆ Zavegepant nasal spray (Zavzpret[®])

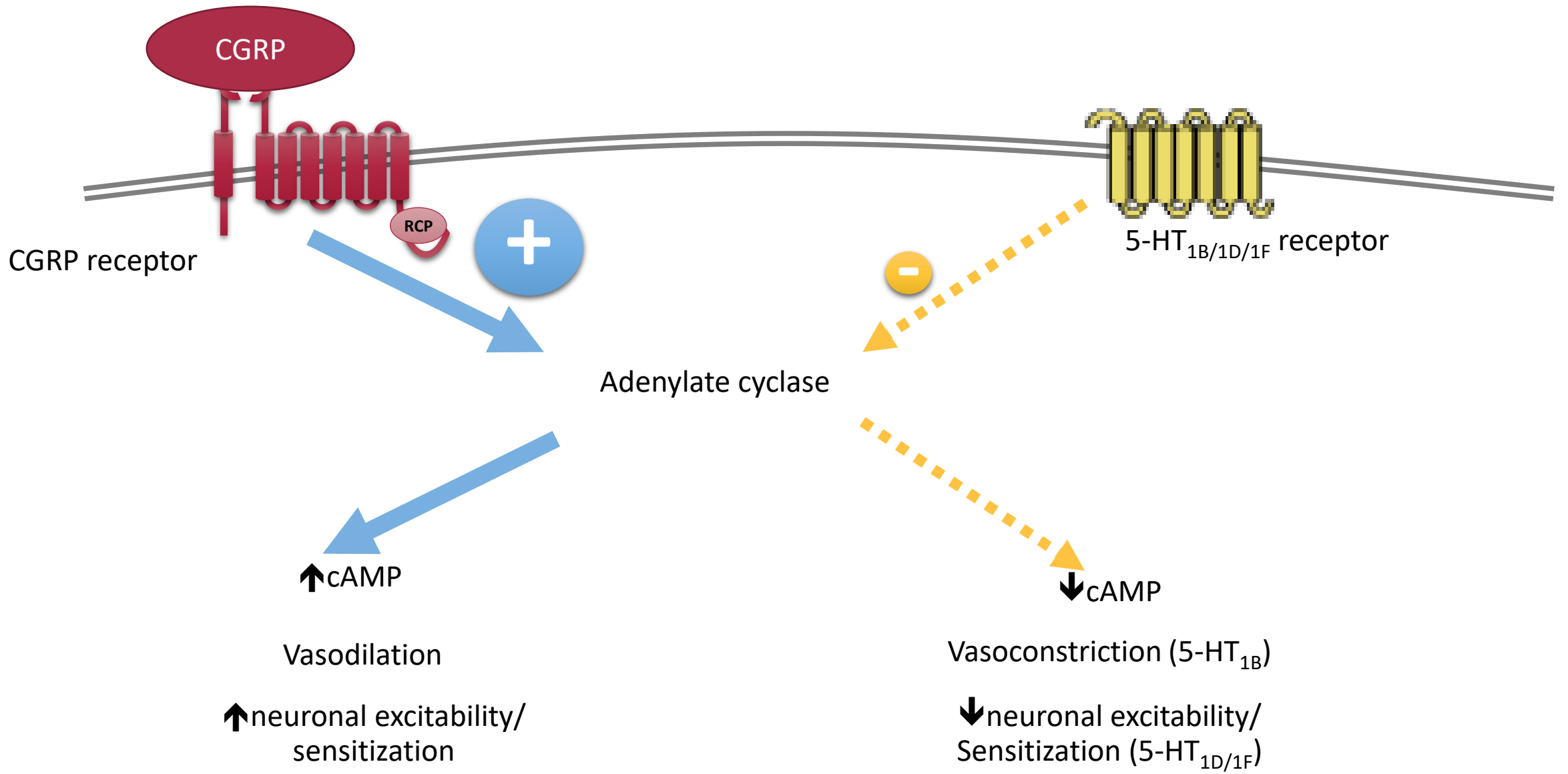
★ Preventive treatment

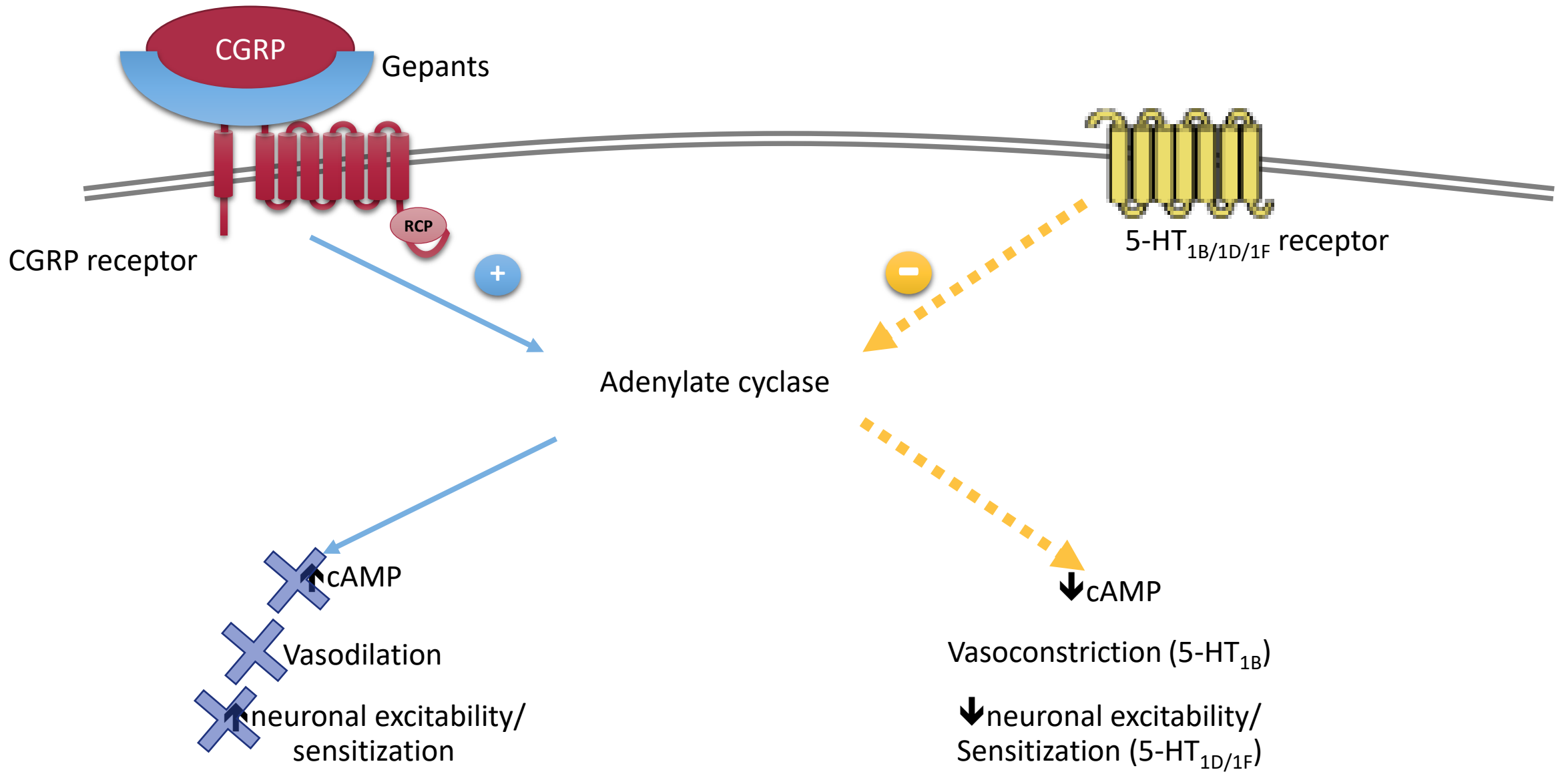
- ◆ Rimegepant orally disintegrating tablet (Nurtec[®] ODT)
- ◆ Atogepant) (Qulipta[®])



CGRP receptor antagonists

- ★ Block CGRP effects
- ★ No direct vasoconstricting side effects
- ★ Minimal adverse effects, in general
- ★ Not induce tolerance or medication overuse headache, unlike ergots/triptans/ditans
- ★ Can be used as abortive and preventive drug





Side effect of gepants

- ★ No severe hepatotoxic at the therapeutic dose ranges
- ★ Generally mild eg. nausea, vomiting
- ★ Long-term use: constipation, blood pressure increase in some cases

Formulations and pharmacokinetics of gepants

	Atogepant	Rimegepant	Ubrogepant	Zavegepant
Formulation	Tablet	Orally disintegrating tablet (ODT)	Tablet	Nasal spray
Tmax	1-2	1-1.5	1.5	0.5
T1/2	11	11	5-7	6-7
Effect of high fat meal	↓ AUC, Cmax 20% Tmax unchanged	↓Cmax 42-53% ↓AUC 32-38%	↓Cmax 22% AUC Unchanged Tmax delayed 2 hr	-
Vd	292	120	350	1774
Protein binding	No	96%	87%	90%
Elimination	CYP3A4	CYP3A4 and minor with CYP2C9	CYP3A4	CYP3A4 and minor with CYP2D6

Dosages of gepants

	Atogepant	Rimegepant	Ubrogepant	Zavegepant
Indication	Preventive treatment of episodic and chronic migraine in adults	- Acute treatment of migraine with or without aura in adults - Preventive treatment of episodic migraine in adults	Acute treatment of migraine with or without aura in adults	Acute treatment of migraine with or without aura in adults
Dosage	- Episodic migraine 10 mg, 30 mg or 60 mg po daily - Chronic migraine 60 mg po daily	- Acute treatment: 75 mg po as needed. Max 75 mg /24 h - Prevention: 75 mg po every other day No established safety of using > 18 doses /30 days	50–100 mg po as needed. Second dose 2 h after the initial dose if needed. Max 200 mg/24 h No established safety of treating > 8 migraines / 30 days	10 mg intranasal No established safety of treating > 8 migraine / 30 days

Migraine preventive medications

Classical

- ★ Anticonvulsants
 - ◇ Topiramate, valproate, gabapentin
- ★ Antidepressants
 - ◇ TCAs, SNRIs
- ★ β -Adrenoceptor blockers
 - ◇ Propranolol
- ★ Calcium channel blockers
 - ◇ Flunarizine, verapamil
- ★ Others
 - ◇ Pizotifen, candesartan

Newer agent targeting CGRP signalling

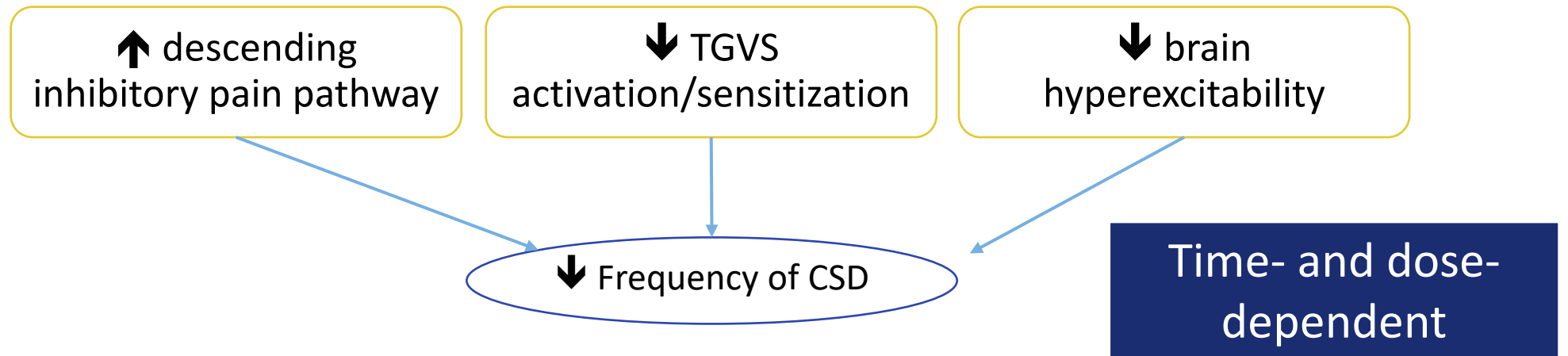
- ★ Monoclonal antibodies
- ★ CGRP receptor antagonist
 - ◇ Rimegepant, atogepant

Medications traditionally used for migraine prevention

	Drug class	Target daily dose	Levels of evidence [*]	Strength of recommendation [†]
Topiramate	Anti-epileptic	100 mg	High	Strong
Propranolol	β-blocker	80–160 mg	High	Strong
Metoprolol	β-blocker	100–200 mg	High	Strong
Amitriptyline	Antidepressant	10–100 mg	High	Strong
Nadolol	β-blocker	80–160 mg	Moderate	Strong
Candesartan	Angiotensin receptor blocker	16 mg	Moderate	Strong
Gabapentin[‡]	Antidepressant	1200–3600 mg	Moderate	Strong
Divalproex sodium	Anti-epileptic	500–1500 mg	High	Weak
Flunarizine	Calcium channel blocker	10 mg	High	Weak
Pizotifen	Serotonin antagonist	1.5–4.0 mg	High	Weak
Venlafaxine	Antidepressant	150 mg	Low	Weak
Lisinopril	Angiotensin-converting enzyme inhibitor	20 mg	Low	Weak
Verapamil	Calcium channel blocker	120–480 mg	Low	Weak
OnabotulinumtoxinA (chronic migraine only)	Neurotoxin	155–195 units	Not applicable	Not applicable

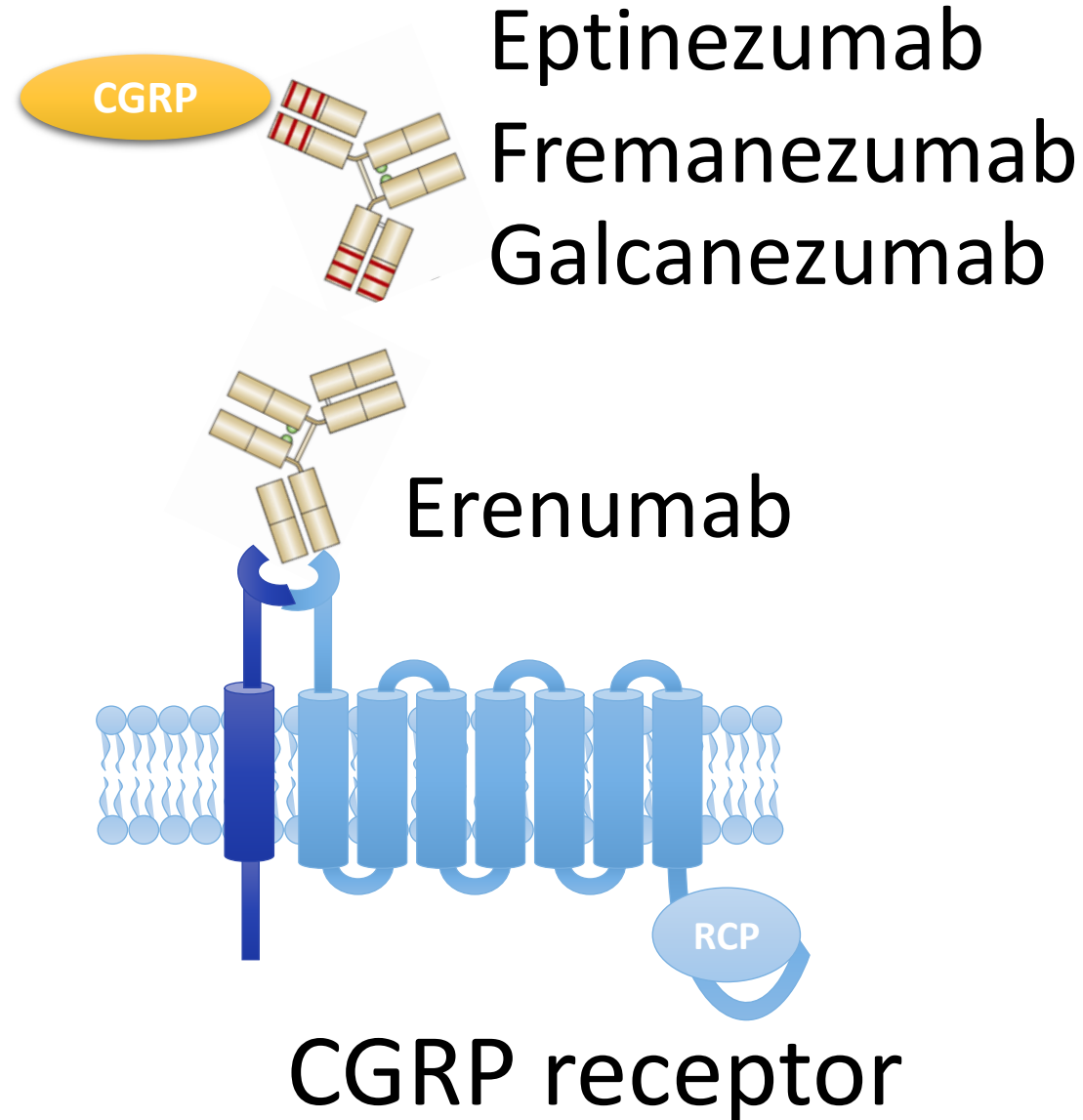
Pharmacologic rationale of migraine preventive medications

- ★ Frequent headache episodes induces TGVS sensitization
- ★ Pharmacological effects of preventive medication



- ★ Acute medications should be provided for breakthrough attacks

Monoclonal antibodies for migraine prophylaxis



Monoclonal antibodies targeting at CGRP/CGRP receptors

Drugs	Erenumab	Fremanezumab	Galcanezumab	Eptinezumab
Synonym	AMG334	TEV48125	LY2951742	ALD403
Tradename	Aimovig®	Ajovy®	Emgality®/ Pontevía®	Vyepti®
Targets	CGRP receptor	CGRP- α and β	CGRP- α and β	CGRP- α and β
mAb types	Human 100%, IgG ₂	Humanised > 95% , IgG ₂	Humanised > 90% , IgG ₄	Humanised, > 90% IgG ₁
Route of admin	SC	SC	SC	IV



Modified from Headache 2018;58:33-47.

Monoclonal antibodies targeting at CGRP/CGRP receptor

Drugs	Erenumab	Fremanezumab	Galcanezumab	Eptinezumab
Route of administration	SC	SC	SC	IV
Tmax (day)	6	5-7	5	n/a
Vd (L)	3.86	6	7.3	n/a
T _{1/2} (days)	21	40-48	28	31
Indication				
• Prevention of EM/CM	✓	✓	✓	✓
• Prevention of eCH			✓	
Dosage and administration	70-140 mg monthly	225mg monthly or 675 mg (225 mg x3) every 3 months	Migraine: 240 mg initially, then 120 mg monthly Cluster: 300 mg at onset then monthly until the end of the cluster period	1000 mg every 3 months

EM=episodic migraine, CM=chronic migraine, eCH= episodic cluster headache

Comparison of mAbs and current oral migraine preventive drugs

mAbs		Current oral migraine preventive drugs
Selectivity	High	Low
Administration	IM/IV	Oral
T1/2	Several weeks to months	<24 hrs
Frequency of admin	1-3 months	≥ 1/day
Drug elimination	Proteolysis	Cytochrome P450 e.g. CYP2D6 (A, P) Phase II enzyme (T)
DDI PK potential	Low to none	High
Dosing titration	No need	Need
Onset of prophylaxis	Rapid, within a week	Slow, several weeks to months
Known ADRs	Injection site pain Long-term clinical significant side effects still unknown	Weight gain (A, V) Weight loss, paresthesia, slow thinking (T) Depressed mood, impotence (P) Sedation, fatigue (A, P, T, V) Xerostomia, constipation and anticholinergic effects (A) Reduced seizure threshold (A)
Immunogenicity	Yes	No (?)

A=amitriptyline, P=propranolol, T=topiramate, V=valproate